

1 Detecting causal models with Mendelian
2 randomization for genomic trios with many
3 confounding variables

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13 Abstract

14 The principle of Mendelian randomization (MR) makes it possible to infer the potential causal
15 relationship between two biological variables (e.g., expression of two genes, or between two
16 molecular phenotypes such as gene expression and DNA methylation), using a genetic variant
17 as an instrumental variable and thus forming a trio. Inference is often influenced by many other
18 variables (such as other genes and demographic variables). Being able to properly account for
19 confounding is therefore essential to causal inference. However, methods that can deal with
20 many confounding variables are limited. Here, we introduce a regression-based method under
21 this principle, namely MR for Genomic Networks in trios (MRGNtrio), to detect diverse causal
22 models for genomic trios using individual-level data while accounting for many confounding
23 variables. We demonstrate the inference power and computational efficiency through exten-
24 sive simulation and application to examine how a single nucleotide polymorphism potentially
25 regulates its cis- and trans-target genes.

26 1 Introduction

27 An important goal in biology is to understand how genes regulate one another and how genetic
28 variation influences gene expression [1, 2, 3, 4]. Gene regulation involves complex networks of
29 high dimensions, which may be represented by directed networks where directed edges repre-
30 sent causal (or regulatory) relationships. Although the ultimate goal in biology is to reconstruct
31 gene regulatory networks at the genome level, it is often of interest to understand the causal re-
32 lationship between a molecular phenotype (e.g. the expression level) of two genes, or between
33 two molecular phenotypes (e.g., between gene expression and DNA methylation), which may
34 be influenced by many other variables such as other genes, other biological processes, demo-
35 graphic variables, etc.

36 To learn the (directed) causal relationship between two phenotypes, we can apply the prin-

37 ciple of Mendelian randomization (PMR) and leverage the information in genetic variants. The
 38 PMR assumes that genotypes in a natural population are randomly assigned, thus observational
 39 omics data can be viewed as being collected from a randomized experiment [5, 6]. This implies
 40 that an edge connecting a genotype to a phenotype will always point to the phenotype, but not
 41 the other way around [7, 8]. As a result, this constraint on edge direction makes it possible to
 42 break the symmetry of association and draw inferences about the regulatory structure among
 43 phenotypes [7, 9, 5].

44 Many methods have been developed under the PMR [10, 11, 12, 13, 7], although they often
 45 are interested in the mediation model $V \rightarrow T_1 \rightarrow T_2$, where V is the genetic variant, T_1 and T_2
 46 are the variables of interest. In this model, the mediator T_1 relays the regulatory information
 47 from V to the outcome T_2 ; for example, a genetic variant regulates the expression level of
 48 a transcription factor, and its reduced expression leads to its downstream target gene being
 49 under-expressed. However, other causal relationships may exist among the three variables.

50 Previously [7] developed MRPC, a method that goes beyond mediation and infers five
 51 basic causal models for a trio that are mutually exclusive (Figure 1) [7, 14]. They include a
 52 null model ($V \rightarrow T_1$ and no relationship between T_1 and T_2 , denoted M0) and four non-null
 53 models: mediation (denoted M1), v-structure (or reverse causation; $V \rightarrow T_1 \leftarrow T_2$, denoted
 54 M2), conditional independence (or horizontal pleiotropy; $T_1 \leftarrow V \rightarrow T_2$, denoted M3), and a
 55 fully connected model ($T_1 \leftarrow V \rightarrow T_2$ and $T_1 \leftrightarrow T_2$, denoted M4).

56 The true model behind M4 may be directed: the edge between T_1 and T_2 is $T_1 \rightarrow T_2$ or
 57 $T_1 \leftarrow T_2$. However, these two directions have the same likelihood and are therefore Markov
 58 equivalent. Without additional information, they are indistinguishable and only an undirected
 59 edge can be inferred.

60 The PMR and the five basic causal models require that no confounding variables can ex-
 61 plain away the directed edge between T_1 and T_2 , since this edge is our main interest. However,
 62 as mentioned earlier, many variables can influence the inference due to the complexity of gene

63 regulation. A confounding variable may influence the inference in different ways. [15] pro-
64 poses three types of confounding for the $T_1 - T_2$ relationship: as a confounder, an intermediate
65 variable, or a common child to T_1 and T_2 . Specifically, a confounder is a common parent to
66 both T_1 and T_2 and can induce a false edge between T_1 and T_2 when unaccounted for. This is
67 similar to T_1 and T_2 being marginally dependent in M3.

68 An intermediate variable acts like a mediator between T_1 and T_2 and can potentially explain
69 away a true $T_1 \rightarrow T_2$ edge when this intermediate variable is accounted for in the inference,
70 leading to a false negative conclusion. By contrast, a common child can induce a false edge
71 between T_1 and T_2 when accounted for, leading to a false positive conclusion. This is because
72 when the common child is accounted for in the inference, T_1 and T_2 become conditionally
73 dependent, similar to their relationship in M2.

74 Accounting for confounding variables is therefore crucial in causal network inference [16,
75 15, 17, 18, 19]. Two basic approaches have emerged to address latent confounding between an
76 exposure and outcome. In the first approach, candidate confounding variables are incorporated
77 into the graph as additional nodes in the network and also treated in the same way as other
78 nodes; our MRPC method and several other methods take this approach [7, 18]. However,
79 it is computationally intensive to include many confounding variables, as the computational
80 complexity for graph inference grows super exponentially with the size of the graph. The
81 second approach focuses on identifying and accounting for confounders [15, 17, 16]. This
82 approach includes all the confounders as a whole in the conditional independence tests, and
83 does not increase the size of the network. For example, we can include the confounders when
84 testing the $T_1 - T_2$ edge in a trio. The trio remains a network of three nodes, but the tests are
85 conditioned on these confounders. However, these methods focus on the mediation model.

86 Herein we present Mendelian Randomization for Genomic Networks in trios (MRGNtrio),
87 a novel machine learning method for inferring the causal model of a trio from individual-level
88 data in the presence of potentially hundreds of confounding variables. MRGNtrio builds on

MRPC and infers the five basic causal models. We take the second approach to confounding described above and focus on accounting for confounders, and not other types of confounding variables. This approach allows for the inclusion of many confounding variables all at once, while maintaining high power and fast computation times. We demonstrate the performance of MRGNtrio and compare to other methods through extensive simulation studies. We further assess this method in our study of the cis- and trans-regulatory modes of genetic variants, using the whole blood data from the Genotype-Tissue Expression (GTEx) Consortium [20].

2 Methods

2.1 The MRGNtrio model

The five causal models for a trio can each be described by a set of marginal and conditional dependence relations (??). Specifically, in the mediation model (M1), any two variables are dependent on each other, but T_2 and V are conditionally independent given T_1 . In the v-structure (M2), T_2 and V are marginally independent but conditionally dependent given T_1 . This is because when the value of T_1 and that of T_2 (or V) are known, the possible values of T_1 's other parent are constrained.

In the conditional independence model (M3), T_1 and T_2 are marginally dependent but conditionally independent given V . In the fully connected model (M4), all pairs of variables are marginally and conditionally dependent. By contrast, in the null model (M0), if V is connected to T_1 (or T_2), then the third variable T_2 (or T_1) is marginally and conditionally independent of V and of T_1 (or T_2). These relations indicate that M0, M1 and M3 can be uniquely determined by conditional (in) dependencies, and that M2 and M4 can be distinguishable from M0, M1 and M3, although marginal dependences are needed to further separate M2 and M4.

MRGNtrio leverages this observation and prioritize statistical tests for conditional dependences, which also allows us to account for a set of confounders, denoted by the vector U ,

113 through the following regression:

$$T_1 = \alpha_1 + \beta_{11}V + \beta_{12}T_2 + \mathbf{\Gamma}_1\mathbf{U} + \epsilon_1; \quad (1)$$

$$T_2 = \alpha_2 + \beta_{21}V + \beta_{22}T_1 + \mathbf{\Gamma}_2\mathbf{U} + \epsilon_2, \quad (2)$$

114 where α_i s are the overall means, scalar β_{ij} s and vectors $\mathbf{\Gamma}_i$ s are effect sizes, and ϵ_i s are random
 115 errors that each follow a normal distribution with mean 0. The conditional (in) dependencies
 116 are encoded in the β s. For example, the null hypothesis $H_0 : \beta_{11} = 0$ represents conditional
 117 independence between T_1 and V given T_2 and confounding variables $\mathbf{U}$, which is denoted
 118 as $T_1 \perp\!\!\!\perp V | \{T_2, \mathbf{U}\}$, whereas the $H_0 : \beta_{21} = 0$ represents conditional independence $T_2 \perp\!\!\!\perp$
 119 $V | \{T_1, \mathbf{U}\}$. If we do not reject $H_0 : \beta_{21} = 0$, which means that T_2 is conditionally independent
 120 of V , then both M1 and M0 are possible models. If we further reject $H_0 : \beta_{22} = 0$, which
 121 means that there is still association between T_2 and T_1 , then we infer this trio to follow an M1
 122 model.

123 M2 and M4 are further distinguished by the marginal relationships between the genetic
 124 variant and each phenotype. In the v-structure M2, if V is connected to T_1 (or T_2), then V and
 125 T_2 (or T_1) are marginally independent. In M4, however, V has marginal dependence on both
 126 T_1 and T_2 . The marginal relationship can also be encoded in the coefficient β in the following
 127 univariate regression:

$$T_i = \alpha + \beta_{i1}V + \epsilon, \quad i = 1 \text{ or } 2. \quad (3)$$

128 Marginal independence is represented by $\beta = 0$. When the data is not informative enough
 129 for any of the basic models to be inferred, we categorize the trio as “other”.

130 We perform t tests for the hypotheses on the regression coefficients. We also allow for a
 131 permutation test for coefficients β_{12} and β_{22} when the normality assumption for the regression
 132 and t test is questionable, because our primary interest is the relationship between T_1 and T_2

133 and also because the relationship between V and either T node is easier to infer. Permutation
 134 is done within each level (i.e., genotype here) of the categorical variable V , similar to [15].

135 For example, to test β_{12} in regression (Equation 1), we permute the values of T_1 within each
 136 genotype and calculate the permutation t statistic t_{12}^* ; this procedure removes any dependence
 137 between T_1 and T_2 while preserving the possible dependence between T_1 and V . The collection
 138 of the permutation t statistics generates a null distribution for t_{12} . We compute a two-sided
 139 permutation p -value as

$$p_{12} = 2P \left(Z > \frac{|t_{12}^{obs}| - \mathbb{E}[t_{12}^*]}{\sqrt{\text{Var}(t_{12}^*)}} \right), \quad (4)$$

140 where $\mathbb{E}$ denotes the expected value of t_{12}^* under the null distribution, and Z is the standard
 141 normal random variable.

142 2.2 Confounder selection

143 As part of the MRGNtrio model, we adopt a similar adaptive confounder selection strategy to
 144 that described in [15]. Given a pool of potential confounders, we first exclude variables that
 145 are marginally associated with V . This filtering step is motivated by the assumption that true
 146 confounders are parents of the T nodes and are therefore independent of each other and of
 147 V . In contrast, variables downstream of one or both T nodes—such as common children or
 148 intermediates—may appear associated with V due to collider bias or mediation effects, and
 149 should not be treated as confounders.

150 Next, candidate confounders are selected based on their statistical association with either
 151 T_1 or T_2 . Following the approach of [15], we test whether the regression coefficients from
 152 modeling each candidate confounder on both T_1 and T_2 are simultaneously zero. This step
 153 involves multiple hypothesis testing, and both the initial filtering and confounder selection
 154 stages employ the q -value procedure to control the false discovery rate (FDR) [21].

155 While we retain the same statistical framework as [15], we substantially improve com-
156 putational efficiency in two key ways. First, we parallelized the confounder filtering and
157 regression-based selection hypothesis tests across candidate confounders, allowing for faster
158 evaluation when the pool of confounders is large. Second, we leverage the R package *prop-*
159 *agate* to improve correlation matrix computations by partitioning the problem into smaller,
160 memory-efficient blocks [22], which scales more efficiently to large genomic datasets.

161 **2.3 Simulation study**

162 We examined the performance of MRGNtrio in a wide range of settings and in comparison
163 to similar methods; MRPC [7] and GMAC [15]. We simulated 1,500 trios by sampling from
164 the five causal models with equal probability. This resulted in approximately 300 trios of each
165 model. For each trio, we applied different combinations of effect size between nodes, noise,
166 and frequency of the minor allele of the genetic variant. We fixed the sample size for all trios
167 at $n = 670$, which is the same sample size of the whole blood tissue in the Genotype-Tissue
168 Expression (GTEx) Consortium (version 8) to which we apply our method [20]. We also in-
169 cluded 1-15 confounders (median: 8), one common child and one intermediate variable in each
170 trio so that we can assess confounder selection and the impact of confounder misspecification
171 on causal inference.

172 When analyzing these trios, we assumed that all the confounding variables were available
173 but we did not know which confounding variables were associated with which trio. In our
174 inference, we first performed the confounding variable selection method of [15] to all the con-
175 founding variables, before applying three methods: MRGNtrio, GMAC and MRPC to each trio
176 with the identified confounders. For all these methods, we used the q -value [21, 23] method
177 for FDR control in confounder selection (denoted CSFDR). For MRGNtrio, we also performed
178 confounder selection without the FDR control (denoted CSnoFDR) (see “Application of causal
179 inference methods”). We further simulated another set of 300 trios with 15–50 confounders

180 (median: 34), representing a scenario with moderate to to extreme genetic confounding. The
 181 candidate confounder pool trios in this scenario contained about 11,000 variables.

182 We simulate genetic trios consisting of a bi-allelic genetic loci with three representative
 183 genotypes and two expression target genes under the five causal structures (Equation 5). Specif-
 184 ically, for models M0 - M2 which each have two possible configurations, we simulate M0 as
 185 $V \rightarrow T_1$; no relationship between T_1 and T_2 , M1 as $V \rightarrow T_1 \rightarrow T_2$, and M2 as $V \rightarrow T_1 \leftarrow T_2$.
 186 Furthermore, since the undirected edge between T_1 and T_2 in model M4 can be represented by
 187 either $T_1 \leftarrow V \rightarrow T_2$; $T_1 \rightarrow T_2$ or $T_1 \leftarrow V \rightarrow T_2$; $T_2 \rightarrow T_1$ we restrict all simulated M4 trios
 188 to the former configuration for simplicity.

189 For all trios, we vary the minor allele frequency, signal strengths of each effect represented
 190 by the coefficients in (5), and the dispersion of the residuals represented by σ . Specifically, we
 191 fix the effect from the genetic variant to be the same for all edges with the genetic variant (i.e
 192 $V \rightarrow T_1$ or $V \rightarrow T_2$), and sample the strength of genetic effect uniformly between 0.5 and 1.5
 193 [15].

194 Similarly, we fix the effect between the genes and sample the strength of the effect between
 195 the genes (i.e the $T_1 - T_2$ edge) uniformly between 0.5 and 1. Lastly, We simulate the strength
 196 of the residual standard deviation σ by scaling the effect of the $T_1 - T_2$ edge such that $\sigma = c\beta$.
 197 We sample the scaling effect c uniformly between 0.3 and 1.5

198 We utilize the topological ordering of the graph of each trio to determine the order in which
 199 nodes are generated. Under this framework, the eQTL is simulated first from a multinomial
 200 distribution under Hardy-Weinberg equilibrium:

$$Pr(V_i = 0) = (1 - \theta)^2; Pr(V_i = 1) = 2\theta(1 - \theta); Pr(V_i = 2) = \theta^2, \quad (5)$$

201 where θ represents the frequency of the minor allele of the genetic variant and V_i is the
 202 genotype of the i^{th} individual. We simulate the minor allele frequency uniformly between 0.01
 203 and 0.5 so that approximately 10% of simulated trios have a "rare" allele.

For the remaining nodes, we simulate each gene or confounding variable Y_j as coming from a normal distribution centered at the linear combination of its parent nodes $S = \mathbf{Parents}(Y_j)$ and with the intercept at the origin (Equation 6). The set of parents S may include the genetic variant, the other gene, confounding variables, a combination, or it may be the empty set.

$$Y_j \sim N \left(\beta_0 + \sum_{i \in S} \beta_i Y_i, \sigma^2 \right) \quad (6)$$

We consider three types of latent confounding variables that may be present for a trio; confounders, intermediate variables, and common child variables (Supplementary Figure 1) [15]. We consider only confounding variables that impact the association between target traits of the genetic variant (T_1 and T_2), as it has been previously established that there are likely few variables that confound the relationship between a genetic variant and its target molecular traits [5, 24].

All trios in our simulation are simulated to have a number of latent confounding variables that must be identified through statistical inference. Specifically, each trio is simulated to have a varying number of confounders, one common child variable, and one intermediate variable. All confounding variable effects are simulated independently from a uniform distribution. Following [15], we set confounder and intermediate variable effects to range between 0.15 and 0.5 and common child effects between 0.5 and 1.5. Additionally, we allow all confounding variable effects a 50% probability for the effect to be negative.

Performance metrics

We summarized the inference results by averaging the performance metrics across all simulated trios using three sets of measures (see “Assessing performance on simulated trios”). First, we calculated recall and precision to assess whether the causal model type (i.e M0, M1, etc.) is

225 correctly inferred for each trio computed as:

$$\text{Recall}_{\text{model}} = \frac{\text{Number of correctly inferred trios}}{\text{Total number of trios of the causal model of interest}}; \quad (7)$$

$$\text{Precision}_{\text{model}} = \frac{\text{Number of correctly inferred trios}}{\text{Total number of trios inferred to be of the causal model of interest}}. \quad (8)$$

226 This performance measure tends to be conservative as a good performance requires the entire
227 trio to be inferred correctly.

228 Second, we calculated recall and precision based on individual edges in a trio. Similar to
229 our earlier work [7], we assigned a score of 0.5 if a true edge is inferred to be present, and
230 another score of 0.5 if the inferred edge has the correct direction. We summarize the edge-
231 based performance across all trios by averaging the individual precision and recall scores:

$$\text{Recall}_{\text{edges}} = \frac{\text{Number of correct edges in inferred graph}}{\text{Total edges in true graph}}; \quad (9)$$

$$\text{Precision}_{\text{edges}} = \frac{\text{Number of correct edges In inferred graph}}{\text{Total edges in inferred graph}}. \quad (10)$$

232 Lastly, we calculated recall and precision based on individual edges in a trio (Equation 9-
233 10). We assigned a score of 0.5 if a true edge is inferred to be present, and another score of
234 0.5 if the inferred edge has the correct direction. We summarize the edge-based performance
235 across all trios by averaging the individual precision and recall scores.

236 Lastly, we incorporate a performance measure strictly for the presence/absence of the edge

237 between the genes ($T_1 - T_2$ edge) to allow for direct comparison of our method with GMAC:

$$\text{Recall}_{T_1 - T_2} = \frac{\text{Number of trios with a correctly identified } T_1 - T_2 \text{ edge}}{\text{Total trios with a true } T_1 - T_2 \text{ edge}}; \quad (11)$$

$$\text{Precision}_{T_1 - T_2} = \frac{\text{Number of trios with a correctly identified } T_1 - T_2 \text{ edge}}{\text{Total trios with an inferred } T_1 - T_2 \text{ edge}}. \quad (12)$$

238 This metric confines the five causal models to two classes: models with $T_1 - T_2$ edge present
 239 (M1, M2, M4) and models with $T_1 - T_2$ edge absent (M0, M3).

240 **2.4 Application of causal inference methods**

241 We compare our MRGNtrio method to two related PMR-based causal inference methods: Ge-
 242 nomic Mediation analysis with Adaptive Confounding adjustment (GMAC) [15] and Mendelian
 243 Randomization PC (MRPC) [7]. Unlike MRGNtrio and MRPC, GMAC is a mediation - based
 244 method that is restricted to inferring only the causal edge between T_1 and T_2 . To that end, we
 245 only compare MRGNtrio and MRPC to GMAC on the performance for predicting the $T_1 - T_2$
 246 edge. We set the number of permutations to 1000 and the test threshold for all marginal and
 247 conditional tests to be $\alpha = 0.01$ which is approximately a Bonferroni correction. For MRPC,
 248 we use the ADDIS FDR control, Gaussian CI test, and set the FDR to 5%. For MRGNtrio,
 249 we employ the permutation test whenever the minor allele is observed in less than 5% of sam-
 250 ples. GMAC, however, applies the permutation test described in to all trios regardless of allele
 251 frequency. For both methods, we set the number of permutations to 1000.

252 Following the approach of [25], we run GMAC twice on all trios, first with T_1 as the
 253 mediator and then with T_2 as the “mediator” and take significant mediation at the $\alpha < 0.05$
 254 and $\alpha < 0.01$ levels. To give GMAC the best chance at predicting the $T_1 - T_2$ edge across
 255 all causal models, we combine the inference results from the tests with T_1 as the mediator
 256 with the tests with T_2 as a mediator. Combining the inference from using each gene as the

mediator establishes four possible results: positive with T_1 as the mediator, positive with T_2 as the mediator, both positive, or both negative. To afford GMAC greater power across the five models, we take any of the aforementioned positive cases for mediation as an indication of the existence of the $T_1 - T_2$ edge.

To emulate a real statistical analysis, we include confounding variable selection as a preliminary step to inference. The confounder selection method of [15] requires the input of a pool of potential confounding variables. In simulation, we construct this pool of potential confounding variables by aggregating the true confounding variables from all simulated trios together (i.e. all confounders, intermediate and common child variables). We control the false discovery rate (FDR) for filtering common child and intermediate variables from the candidate pool at the 10% level. Furthermore we consider two settings for selecting confounders from the pool post-filtering (i) selecting confounders with no FDR correction and significant confounders selected at the $\alpha < 0.01$ (denoted CSnoFDR) and (ii) using the qvalue method to control the FDR of selected confounders at the 5% level (denoted CSplusFDR). The latter setting is the default for selecting confounders using the GMAC software.

2.5 Application to GTEx data

The GTEx consortium has mapped a large number of eQTLs and their associated cis target genes (cis-eQTLs and cis-target genes) for most genes in the human genome across many human tissue types [20]. Recent studies have focused on identifying cis-target genes that mediate the relationship between the eQTL and its associated trans-target genes [15]. We previously have applied the MRPC method to all the GTEx tissues to detect diverse causal models for such trios: an eQTL and its associated cis- and trans-target genes, where the genes are protein coding genes or long non-coding RNAs (lncRNAs) [25].

We implement the preprocessing and normalization pipeline from Section 2.2 to construct trios for WholeBlood tissue from version 8 of GTEx. Each trio consists of an eQTL and its

highest association “cis” and “trans” target genes. We define cis genes as genes less than one megabase from the eQTL and trans genes as genes residing more than one megabase from the eQTL or on a different chromosome. As noted in Section 2.2 individual trios may share the same eQTL or the same gene, and the gene may be a cis-gene in one trio but a trans gene in another trio.

In GTEx version 8, the number of available samples for whole blood is 670. After preprocessing, 8,823 trios were identified. To eliminate bias induced by mismapping transcripts, we restricted our final set of trios to protein coding genes or long non-coding RNA genes [15]. This additional filtering step reduces the number of trios to 3,248. Many of eQTLs provided by the GTEx consortium contain missing values. Since the GMAC algorithm is unable to handle missing values, we used multiple correspondence analysis implemented in the *missMDA* package in *R* [26] to impute the missing genotypes of individuals using the collection of unique eQTLs from all GTEx trios analyzed for whole blood.

Confounding variable selection and filtering for GTEx trios

We performed principal component analysis (PCA) on the PEER-normalized, genome-wide gene expression data and used the resulting principal component (PC) scores as the pool of candidate covariates for confounder selection. Each PC summarizes the regulatory effects of a large number of genes across the genome. Because each trio represents a small component of a larger regulatory network, PCs that correlate with a trio may capture potential confounding effects arising from pleiotropy. We performed confounder selection as described in Confounding Variable Selection and applied common-child and intermediate-variable filtering to obtain a final set of confounders for each trio. Following our findings from simulation, we applied MRGNTrio using the CSnoFDR setting and selected confounders at a significance threshold of $\alpha < 0.05$. While the simulation study used a more stringent cutoff of $\alpha < 0.01$, we relaxed this threshold here because the more stringent cutoff produced nearly the same set of confounders

as the CSFDR setting when applied to the whole-blood PC score matrix. For each trio, we also included sex, platform, and PCR batch as additional covariates.

3 Results

3.1 Simulation Results

MRGNtrio improves power and precision in the presence of confounding

MRGNtrio achieves substantially higher recall for the mediation, v-structure, and fully connected models than MRPC does (M1: +82%, M2: +47%, and M4: +25%; (Table 2). All these models contain a $T_1 - T_2$ edge, indicating the sensitivity of MRGNtrio in detecting the $T_1 - T_2$ edge. On the other hand, this sensitivity leads to reduced power to detect the null and conditional independence models (M0: -37%, M3: -38%), where there is no edge between the T nodes. Remarkably, MRGNtrio also achieves equivalent or better precision for nearly all models (M0: +47%, M1: +29%, M2: +48%, M3: +8%, and M4: -17%).

When comparing the performance using edge-level recall and precision, MRGNtrio infers more edges than MRPC while maintaining a high degree of precision. Furthermore, MRGNtrio matches the performance of GMAC for the $T_1 - T_2$ edge (Supplementary Figure 2), as the GMAC test for mediation is a special case of our method. The similar performance also shows that the notion of “mediation” detected by GMAC is broad, while our method is able to distinguish the different models behind the broadly-defined mediation.

MRGNtrio handles many confounding variables

In scenarios with moderate to severe confounding, MRGNtrio and GMAC performed similarly well in recall and maintained consistently high recall for the $T_1 - T_2$ edge, even under substantial confounding, with precision remaining relatively unchanged (Supplementary Figure 2). Additionally, recall and precision across all edges remained high, indicating that MRGNtrio

330 can accurately infer most trio edges, except in cases with many (> 35) confounders, where
331 recall declined but still exceeded that of MRPC Supplementary Figure 3.

332 **The impact of confounder selection on causal network inference**

333 Accurate selection of confounding variables is a critical step preceding causal inference in
334 Mendelian randomization studies, as errors at this stage can propagate to the causal model by
335 omitting informative genetic or environmental covariates. In our simulation study, we com-
336 pared two strategies for selecting confounders from a pool of candidate covariates: CSnoFDR,
337 which selects confounders without false discovery rate (FDR) correction, and CSFDR, which
338 controls the FDR at 5%. We first evaluated both methods based on their ability to identify
339 the true confounders, measured by precision and recall. Overall, both methods achieved recall
340 greater than 90% when the number of true confounders per trio was modest (e.g., fewer than
341 25; see Figure 2). However, CSFDR showed reduced performance when for trios with many
342 true confounders, with average recall to detect true confounders dropping to approximately
343 65%. Conversely, CSnoFDR achieved nearly perfect recall across all simulations, but at the
344 expense of very low precision, indicating that it selected many false confounders in addition to
345 the true ones.

346 When we examined how confounder selection affected causal inference performance, we
347 found no clear relationship between the number of false-positive confounders and MRGNTrio's
348 ability to accurately infer causal edges (Figure 3). In contrast, falsely omitted confounders had
349 a strong negative impact on causal inference performance. Overall, we observed little differ-
350 ence in causal inference performance between the two methods in our simulation study. This
351 may be because only a small number of trios were simulated with a large number of confound-
352 ing variables (e.g., more than 35), limiting our ability to detect any differences between the two
353 methods. Although CSnoFDR is more liberal, it may be preferable in scenarios where many
354 potential covariates are expected for a trio.

Speed comparison of MRGNtrio, MRPC and GMAC

Our analysis involves two major steps: confounder selection and causal network inference. We report the computation time for each step separately. We conducted a comparative analysis of the time required for confounder selection across varying sizes of the confounder covariate pool (100, 500, or 1000 covariates) and different numbers of trios for which confounders needed to be identified (500 or 1000), resulting in a total of six replicates. Notably, in the most intensive case 1000 covariates and 1000 trios, our method accomplished confounder selection within 16 minutes (Supplementary Figure 4). This represents a notable gain in computation efficiency compared to the implementation provided by GMAC, which took approximately 28 minutes. Although our confounder selection procedure is similar to that of [15], our implementation is more efficient due to the use of the R package propagate [22], which enables fast correlation calculations between each trio's target genes and the candidate covariate pool, significantly reducing the time required for confounder selection.

In causal network inference, MRGNtrio is substantially faster than both MRPC and GMAC (Table 3). The median time to analyze a trio using the same set of confounders was 0.0065 seconds for MRGNtrio, compared to 0.3557 seconds for MRPC and 0.7088 seconds for GMAC. This makes MRGNtrio over 50 times faster than MRPC and more than 100 times faster than GMAC, on average.

The extended inference time for MRPC is largely due to the ADDIS method used for false discovery rate (FDR) control, which was particularly slow for some trios. When MRPC is applied using the more conservative LOND method for FDR control, its runtime becomes comparable to that of MRGNtrio. For GMAC, we note that our comparison used the default single-threaded implementation provided in its R package. Enabling the parallelization option available in GMAC can substantially reduce its computational time.

Confounding contributes to an inflated type I error rate

In our simulation study, we observed significantly higher rates of false positive mediation for the GMAC method than those reported by [15]. In their simulations under the null model, [15] reported type I error rates (false mediation) consistent with the significance level used to test for mediation. However, in our simulations, we observed type I error rates for the $T_1 - T_2$ edge under the null model to be as high as 75% for both MRGNtrio and GMAC at the $\alpha < 0.01$ significance cutoff. We attribute these differences to several distinctions between our simulation framework and theirs.

First, we simulated a substantially larger number of confounders for each trio (1–50 across all trios in our simulations, compared to 1–2 in [15]). Additionally, to construct the pool of confounders used for selection, we aggregated all confounding variables from all simulated trios, resulting in a large candidate pool of more than 10,000 variables. We also simulated each trio to have one common child and one intermediate variable, which were included in the candidate covariate pool. In contrast, [15] simulated a much smaller pool of 350 confounders, yielding an average of approximately two confounders per trio, with many trios having no confounders. Furthermore, their simulations treated each type of variable (confounder, common child, intermediate variable) separately, whereas our simulations included all three types simultaneously.

To validate the error rates reported by [15], we incorporated their simulation design into our framework to examine the effect of confounders on type I error (false mediation) under the null model. We simulated 100 additional trios for each of seven scenarios containing 2, 4, 6, 8, 10, 12, or 15 confounders, excluding common child and intermediate variables, and applied GMAC with CSFDR confounder selection using our approach of aggregating covariates across simulated trios to construct the confounding variable pool. The scenario with two confounders is comparable to the null-model simulations reported by [15], while the remaining scenarios align with our simulation settings. We successfully replicated the type I error rates

reported by [15] in the two-confounder scenario (Supplementary Figure 9), whereas all other scenarios showed increasingly inflated rates of false mediation that scaled with the number of confounders and the size of the covariate pool. These results demonstrate that substantial confounding leads to markedly inflated false positive mediation and further support the low power to detect the null and conditional independence models observed in our simulation study.

3.2 Application results

We applied MRGNTrio combined with the CSnoFDR confounder selection procedure to 3,248 eQTL trios for whole blood in GTEx. We compared the results from MRGNTrio with those from MRPC (using the ADDIS method to control the overall FDR of conditional tests) and GMAC [7, 27]. Principal component analysis (PCA) was performed across all genes in whole blood to derive 670 principal components, which served as the pool of candidate confounding variables. For each trio, we identified a median of 51 confounders (SD = 6.38) (Figure 4).

After applying MRGNTrio, 911 trios were classified as the null model, 91 as mediation, 10 as v-structures, 2,062 as conditionally independent, and 104 as fully connected. The remaining 70 trios were categorized as “other.” We compared these results with our previous work using MRPC [25], and found that MRGNTrio identified more than twice as many cases of causal mediation and 173 additional cases of conditional independence. Similar to MRPC, MRGNTrio also distinguishes two types of mediation: (1) cis-gene mediation $V \rightarrow T_1 \rightarrow T_2$, where the cis-gene mediates the signal from genetic variant to the trans-gene, and (2) trans-gene mediation $V \rightarrow T_2 \rightarrow T_1$, where the trans-gene acts as the mediator between the genetic variant and the cis-gene. Among the 91 trios identified as mediation by MRGNTrio, 69 (75.5%) had the cis-gene as the mediator and 22 (24.5%) had the trans-gene as the mediator (Supplementary Table 3). Compared to MRPC, MRGNTrio identified not only more mediation trios overall, but also a higher proportion of cis-gene mediation. However, only 8 cis-gene mediation trios and 6 trans-gene mediation trios were identified by both methods.

430 We further investigated trios where MRGNTrio inferred mediation but MRPC did not and
431 found that these discrepancies were primarily driven by differences in the sets of selected con-
432 founders (see Supplementary Text 1; Examples 1–2). In two illustrative cases, MRGNTrio
433 selected 40 and 43 confounders, whereas MRPC selected only three and none. When we reran
434 MRGNTrio using the same confounder set and inference model as MRPC, both methods pro-
435 duced identical results, confirming that accurate identification of confounders is essential for
436 consistent causal inference.

437 **Permutation testing causes conservative model shifts among GTEx trios**

438 Comparing MRGNTrio and GMAC results on GTEx trios, we observed minimal overlap in the
439 identified cases of mediation, similar to our comparison with MRPC. Of the 57 trios identified
440 as mediation by GMAC, only four (two cis-gene mediated and two trans-gene mediated) were
441 also inferred as mediation by MRGNTrio. Two key factors likely contribute to this difference:
442 (i) GMAC uses the CSFDR method for confounder selection, whereas MRGNTrio was applied
443 using CSnoFDR, and (ii) GMAC applies a permutation test to all trios, whereas MRGNTrio
444 applies it only to trios with rare alleles.

445 Regarding confounder selection, GMAC adjusted for a median of 30 confounders, whereas
446 the more liberal CSnoFDR method used for MRGNTrio adjusted for a median of 51. This
447 highlights a substantial difference in the inference models applied by the two methods.

448 To assess the effect of the permutation test, we applied MRGNTrio using permutation-
449 based conditional independence tests across all GTEx trios. This changed the inferred model
450 for 89 of the 3,248 trios. Trios affected by permutation had significantly lower minor allele
451 frequencies ($\leq 14\%$) compared to those unaffected ($t = -7.69$, $p = 7.2 \times 10^{-13}$; Supple-
452 mentary Figure 5). Approximately 97% of permutation-induced changes were conservative,
453 typically involving edge removal or reclassification as “Other” (see Supplementary Text 1; Ex-
454 amples 3–4). Similar conservative behavior was observed in our simulation study, although no

455 significant correlation with allele frequency was detected (Supplementary Figure 6).

456 Finally, we examined the impact of allele frequency on gene expression in GTEx trios.
457 Low-frequency alleles were generally associated with right-skewed trans-gene expression, while
458 cis-gene expression remained approximately normal (Supplementary Figure 7). Trios affected
459 by permutation exhibited higher skewness and kurtosis, reflecting residual non-normality de-
460 spite normalization efforts. This non-normality was typically driven by a small number of
461 individuals with extreme trans-gene expression, as illustrated in Supplementary Figure 8. In
462 these cases, the permutation test mitigated the influence of such outliers, suggesting it serves
463 as an important conservative measure for trios with rare alleles.

464 **MRCNtrio is robust to model misspecification in causal mediation**

465 Based on our simulation and application findings, we conducted a secondary analysis to in-
466 vestigate how model misspecification affects the detection of causal mediation. In particular,
467 we examined how inaccuracies in the inference model, particularly in terms of confounding
468 variables, influence the power to detect mediation on the $T_1 - T_2$ edge. We base our simula-
469 tions on 192 Whole Blood trios that were positive for mediation under the GMAC mediation
470 test. We consider four simulation scenarios: (i) when the inference model includes a few false
471 positive confounders (denoted LTMS) (ii) when the inference model is missing a few true con-
472 founders with weak effects (denoted STMS) (iii) when the inference model is correct (denoted
473 CMS) (iv) when the inference model is missing many true confounders with moderate to strong
474 effects (SIMS)(See Supplementary Text 1 for details and methods).

475 Our simulations show that when the inference model contains most of the influential con-
476 founding variables, the correct inference for the $T_1 - T_2$ edge can typically be recovered (Ta-
477 ble 4). The first two simulation scenarios represent relatively minor changes in the confounding
478 variables used in the inference model, either by omitting or including weak confounders. In
479 these scenarios, correct inference could typically be achieved, with the exception of a small

subset of trios where even the correct model failed to yield accurate results.

The final scenario (SIMS) reflects the most significant discrepancy between the true model that generates the data and model used to infer the causal model. Here, the inference model omitted an average of 28 confounding variables ($SD \approx 7$), many of which had moderate to strong confounding effects. In this scenario, the type II error rate climbed to as high as 30%, underscoring the critical need to retain high-impact confounders for accurate causal inference.

4 Discussion

In summary, we have developed MRGNtrio to improve the power and accuracy of detecting key causal relationships in diverse genomic trios, particularly in the presence of multiple confounding variables. Our method adopts an extended interpretation of the PMR [7] and leverages conditional and marginal hypothesis testing to identify five basic causal trio structures. By coupling MRGNtrio with a relaxed version of the adaptive confounder selection procedure described by [15], our method achieves efficient and precise inference of causal edges without the computational penalty typically associated with high-dimensional confounding.

Our simulations demonstrate that MRGNtrio outperforms comparable methods in detecting key causal edges ($T_1 - T_2$) and mediation models (M1) under realistic settings. A major advantage of our approach is the avoidance of the multiple testing burden; by adjusting for selected confounders within a regression framework, MRGNtrio incorporates numerous confounders without the need to infer a graph larger than three nodes. Furthermore, by integrating a permutation test for the $T_1 - T_2$ edge, our method achieves results comparable to GMAC. Our comparisons suggest that the permutation test offers a conservative measure, reducing the risk of Type I errors in the presence of non-normal data or rare genetic effects.

Our results also underscore the critical importance of accurate confounder selection in causal inference. Omitting key confounders drastically reduces the power to detect mediation, leading to Type II errors and the loss of regulatory insights. Conversely, the inclusion

of false-positive confounders has a negligible impact, particularly when their effects are weak. This suggests that in practical applications, a liberal strategy favoring the inclusion of many potential confounders is preferable. Ultimately, MRGNtrio balances this requirement without excessive computational cost, making it a robust tool for genomic research.

4.1 Algorithmic differences in confounder handling

PC-based methods such as MRPC initialize learning by filtering edges from a fully connected graph structure. After removing edges with weak marginal relationships, the remaining edges are tested sequentially by conditioning on an increasing neighborhood of nodes [7, 28]. Because the number of tests required to determine the existence of an edge is unknown a priori, MRPC utilizes sequential FDR control [29]. Consequently, the testing burden is heavily dependent on graph connectivity, signal strength, and the total number of nodes.

Critically, MRPC treats confounding variables as additional nodes within the graph. When confounders are included, the number of conditional tests increases significantly, and only edges with strong supporting evidence survive. Thus, MRPC inherently infers a larger network than MRGNtrio for the same biological question. Additionally, the adaptive FDR can result in significance thresholds that deviate substantially from the fixed $\alpha = 0.01$ employed by MRGNtrio. As demonstrated by our simulations, this causes MRPC to act more conservatively in the presence of confounding variables.

While our simulations and real data analysis demonstrate the flexibility of MRGNtrio in robustly estimating causal relationships, the method is currently limited to trios containing a genetic variant that acts as an instrumental variable. Similar to MRPC, MRGNtrio relaxes strict instrumental variable assumptions by empirically validating the association between the variant and the phenotype(s). We are currently extending this approach to develop a general MRGN algorithm capable of accommodating larger networks and addressing other classes of confounding variables.

Data access

Data from The Genotype-Tissue Expression (GTEx) Project version 8 were obtained from dbGaP (<http://www.ncbi.nlm.nih.gov/gap>) through accession number phs000424.v8.pht002741.v8. Open-source R packages used in this manuscript include <https://github.com/Jarred6068/MRGN>, MRGNtrio (version 1.3.7.3); <https://github.com/audreyfulab/MRGN>, MRPC (version 3.1.0); <https://cran.r-project.org/web/packages/MRPC/index.html>, and GMAC (version 3.1; <https://cran.r-project.org/web/packages/GMAC/index.html>). Detailed results generated by our analyses are included in the supplementary material.

Competing interest statement

The authors declare no competing interests.

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563 **Author Contributions**

564 J.K and A.Q.F designed the study, J.K developed the tools and analyzed the data, and J.K and
565 A.Q.F wrote the manuscript.

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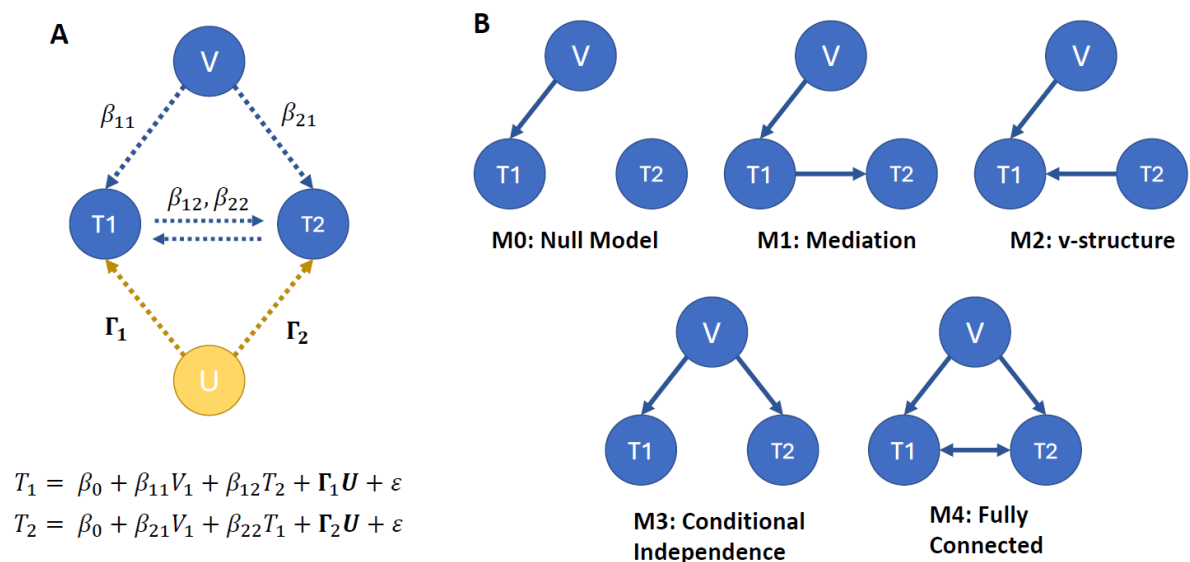


Figure 1: Causal models for a genetic trio. (A) A causal diagram of a genomic trio including a genetic variant (V), two molecular phenotypes (T_1 and T_2), and a set of confounding variables (U). Regression coefficients from Equation 1 and Equation 2 correspond to the directed edges in the graph. (B) The five possible causal models for a trio under the principle of Mendelian randomization (adapted from [7]). All other configurations are categorized as “other.”

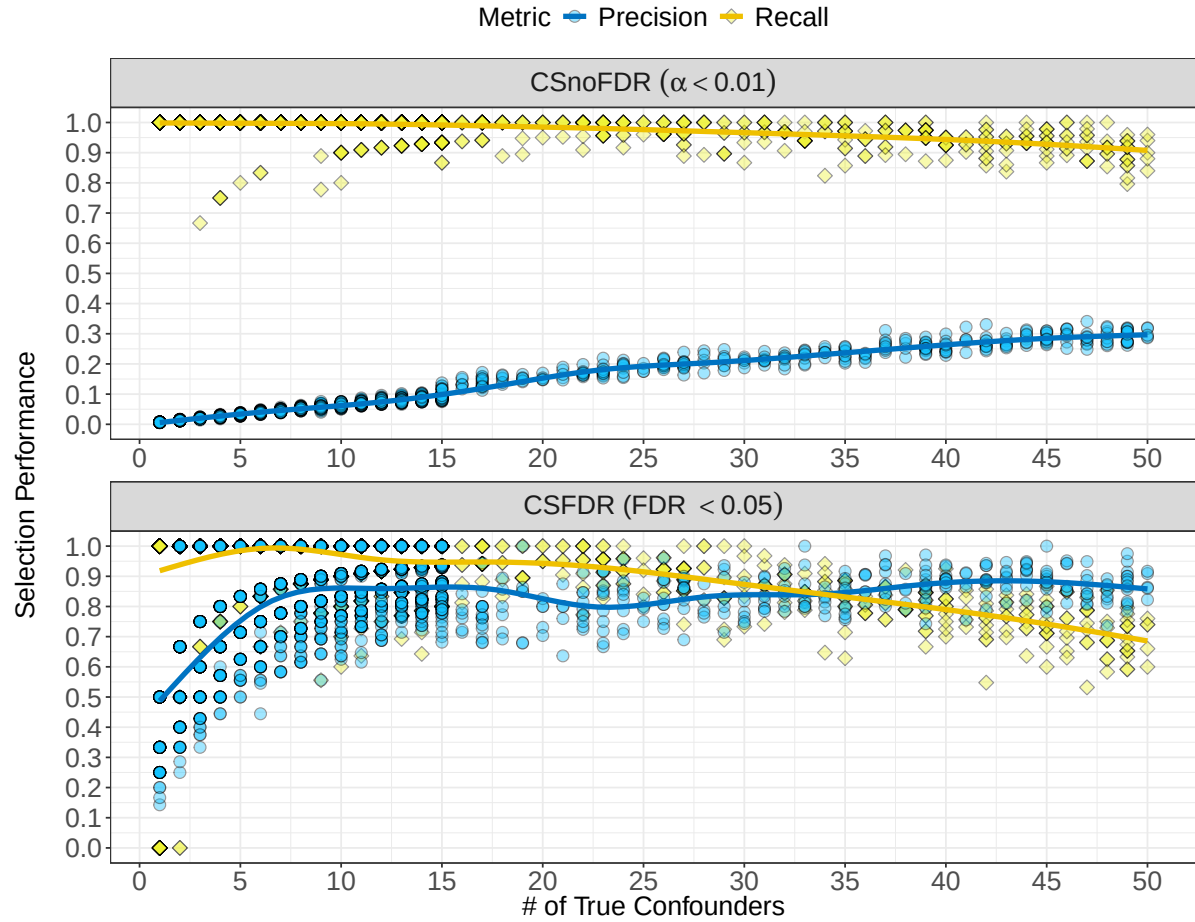


Figure 2: Precision–recall curves comparing confounder selection performance of the CSFDR and CSnoFDR methods. Recall is defined as the proportion of true confounders correctly identified and selected for the inference model while precision is the proportion of true confounders selected out of all confounders selected.

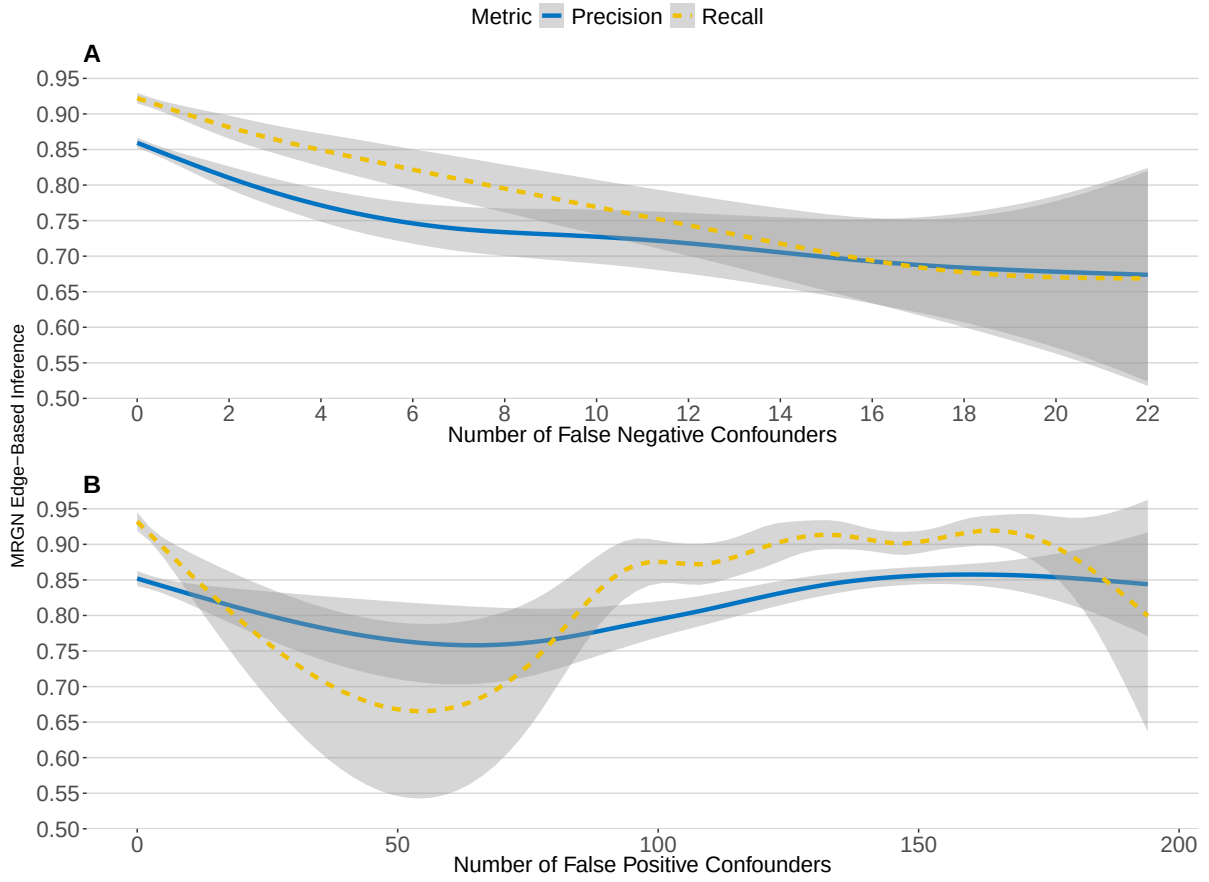


Figure 3: Relationship between confounder selection and causal inference performance measured as precision and recall of edge prediction. (A) Edge-based performance of MRGNTrío on simulated trios as a function of the number of true confounders falsely omitted from the analysis. (B) Edge-based performance of MRGNTrío on simulated trios as a function of the number of falsely included confounders.

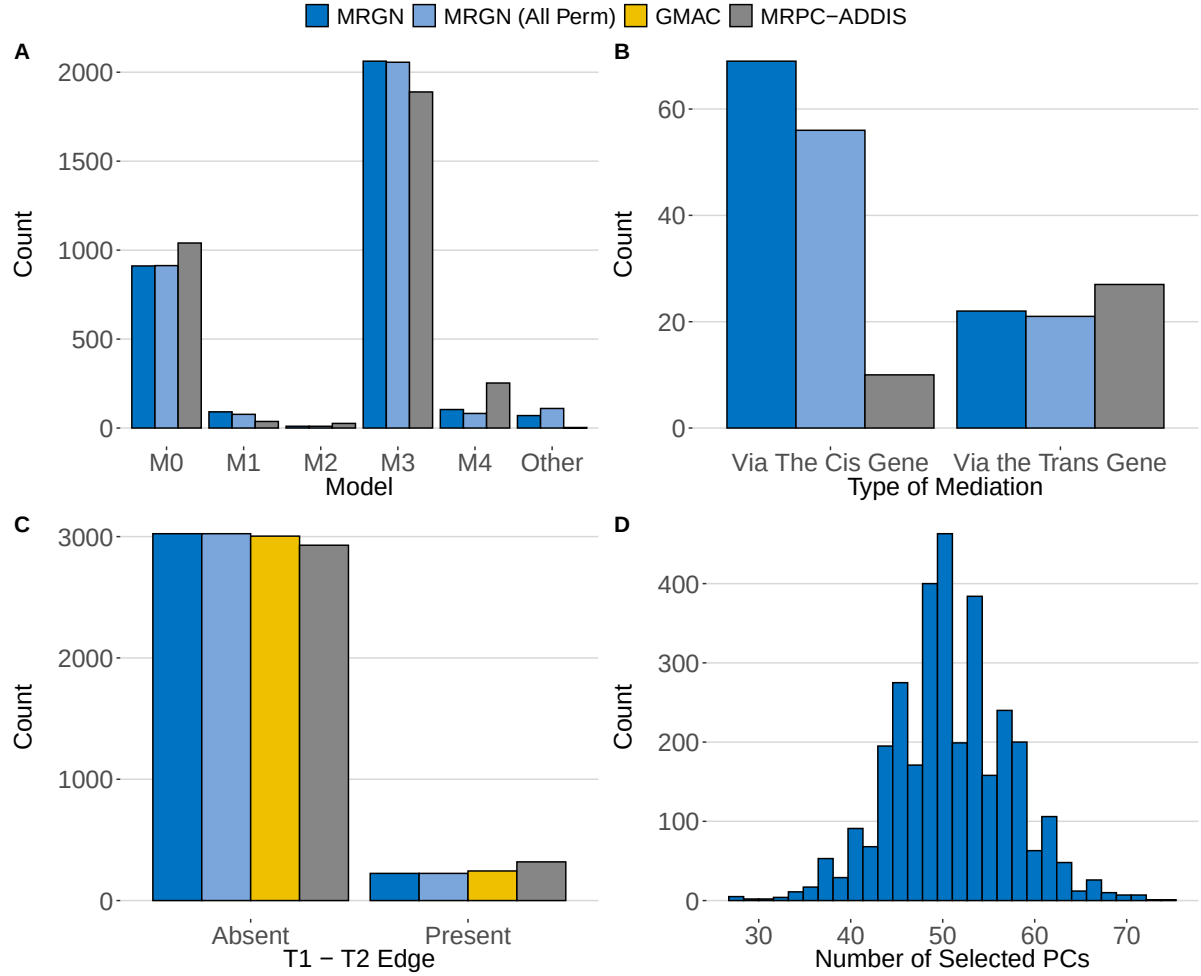


Figure 4: Comparative analysis of MRGNtrio, MRPC [30] and GMAC [15] on GTEx human whole blood trios. We applied MRGNtrio (standard), MRGNtrio with permutation testing (All Perm), GMAC, and MRPC (using ADDIS FDR control) to genomic trios. (A) Count of inferred causal models (M1–M4 and Other) comparing MRGNtrio and MRPC. (B) Comparison of specific mediation pathways (via cis-gene vs. trans-gene) identified by MRGNtrio and GMAC. (C) Total counts of trios where the $T_1 - T_2$ causal edge was inferred as Present or Absent across all four methods[cite: 33, 34, 36]. (D) Histogram displaying the frequency distribution of the number of Principal Components (PCs) selected as confounders by the CSnoFDR method.

Table 1: Conditional and marginal dependence tests used by MRGNTrío. This table summarizes the tests applied to determine which model is supported by the data. The first three models each have two possible configurations. Coefficients β_{11} and β_{12} come from the regression in Equation 1, and β_{21} and β_{22} from the regression in Equation 2; these coefficients are used to assess conditional independence. Marginal independence is tested using the correlations r_{V,T_1} and r_{V,T_2} .

		Conditional Tests				Marginal Tests	
Dependence Relation Conditioned on		$T_1 \perp\!\!\!\perp V$ $T_2, \mathbf{U}$	$T_1 \perp\!\!\!\perp T_2$ $V, \mathbf{U}$	$T_2 \perp\!\!\!\perp V$ $T_1, \mathbf{U}$	$T_2 \perp\!\!\!\perp T_1$ $V, \mathbf{U}$	$V \perp\!\!\!\perp T_1$ –	$V \perp\!\!\!\perp T_2$ –
Model	Configuration	β_{11}	β_{12}	β_{21}	β_{22}	r_{V,T_1}	r_{V,T_2}
M_0	$V \rightarrow T_1$; T_2 is singleton	$\neq 0$	$= 0$	$= 0$	$= 0$	–	–
	$V \rightarrow T_2$; T_1 is singleton	$= 0$	$= 0$	$\neq 0$	$= 0$	–	–
M_1	$V \rightarrow T_1 \rightarrow T_2$	$\neq 0$	$\neq 0$	$= 0$	$\neq 0$	–	–
	$V \rightarrow T_2 \rightarrow T_1$	$= 0$	$\neq 0$	$\neq 0$	$\neq 0$	–	–
M_2	$V \rightarrow T_1 \leftarrow T_2$	$\neq 0$	$\neq 0$	$\neq 0$	$\neq 0$	$\neq 0$	$= 0$
	$V \rightarrow T_2 \leftarrow T_1$	$\neq 0$	$\neq 0$	$\neq 0$	$\neq 0$	$= 0$	$\neq 0$
M_3	$T_1 \leftarrow V \rightarrow T_2$	$\neq 0$	$= 0$	$\neq 0$	$= 0$	–	–
M_4	$T_1 \leftarrow V \rightarrow T_2$; $T_1 \longleftrightarrow T_2$	$\neq 0$	$\neq 0$	$\neq 0$	$\neq 0$	$\neq 0$	$\neq 0$

Table 2: Recall and precision for MRGNtrio and MRPC-ADDIS on simulated trios with confounding variables. The highest precision and recall for each model or edge is highlighted in each row. The first column indicates the performance of MRGNtrio when the true set of confounders is used in inference for comparison. We also consider two selection methods for inferring the confounders for each trio: (i) FDR correction with the qvalue method to control the FDR at 5% (CSFDR) and (ii) no FDR correction and significance taken at the $\alpha < 0.01$ level (CSnoFDR). Additional columns represent performance for MRPC-ADDIS using the CSDFR method for selecting confounders. Rows 1-5 indicate class-based performances for each of the 5 causal models, row 6 gives performance for predicting the $T_1 - T_2$ edge (* indicates similar values were observed for GMAC), and the last row gives the performance across all edges in the network using the weighting scheme described in the Section “Performance metrics”.

Metric	MRGNtrio + GT		MRGNtrio + CSFDR		MRGNtrio + CSnoFDR		MRPC-ADDIS + CSFDR	
	Recall	Prec.	Recall	Prec.	Recall	Prec.	Recall	Prec.
Null Model (M0)	0.562	0.995	0.452	1.000	0.477	0.951	0.836	0.532
Mediation (M1)	0.982	0.516	0.877	0.438	0.883	0.429	0.097	0.147
V-structure (M2)	0.919	0.985	0.892	0.925	0.842	0.968	0.377	0.443
Cond. Indep. (M3)	0.515	0.989	0.283	0.910	0.305	0.872	0.609	0.828
Fully Connect. (M4)	0.600	0.566	0.501	0.425	0.464	0.419	0.217	0.592
All edges	0.945	0.893	0.931	0.864	0.911	0.857	0.770	0.892
$T_1 - T_2$ edge	1.000	0.767	0.994*	0.703*	0.9787	0.7071	0.7884	0.916

Table 3: Inference times for MRGNtrio, MRPC-ADDIS, and GMAC on simulated trios. MRGNtrio shows a substantially faster median inference time per trio compared to both MRPC-ADDIS and GMAC, demonstrating its computational efficiency in the presence of confounding.

Method	Median time (s)	IQR (s)
MRGNtrio	0.0105	0.0052
MRPC-ADDIS	0.4601	9.0117
GMAC	1.5373	1.4831

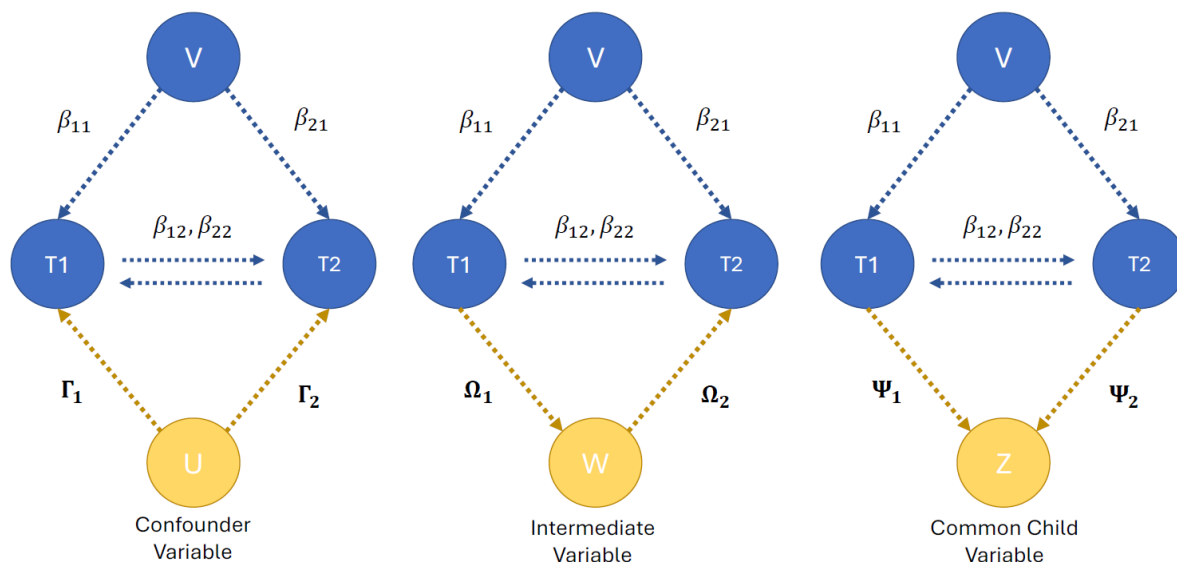
Table 4: A summary of type II error rates for MRGNtrio under model misspecification.

This table reports the type I and type II error rates under four simulation scenarios investigating model misspecification. Small True Model Simulation (STMS): The trans gene is simulated using only the most significant confounders, while the inference model includes additional, less significant confounders, representing model overspecification. Large True Model Simulation (LTMS): The trans gene is simulated with a larger set of confounders, but the inference model omits weaker confounders, representing model underspecification. Correct Model Simulation (CMS) - The same model is used for both simulation and inference, assessing the stability of the permutation test when the model is correctly specified. Small Inference Model Simulation (SIMS) - The trans gene is generated using a full model, but inference is performed with only a small subset of top confounders.

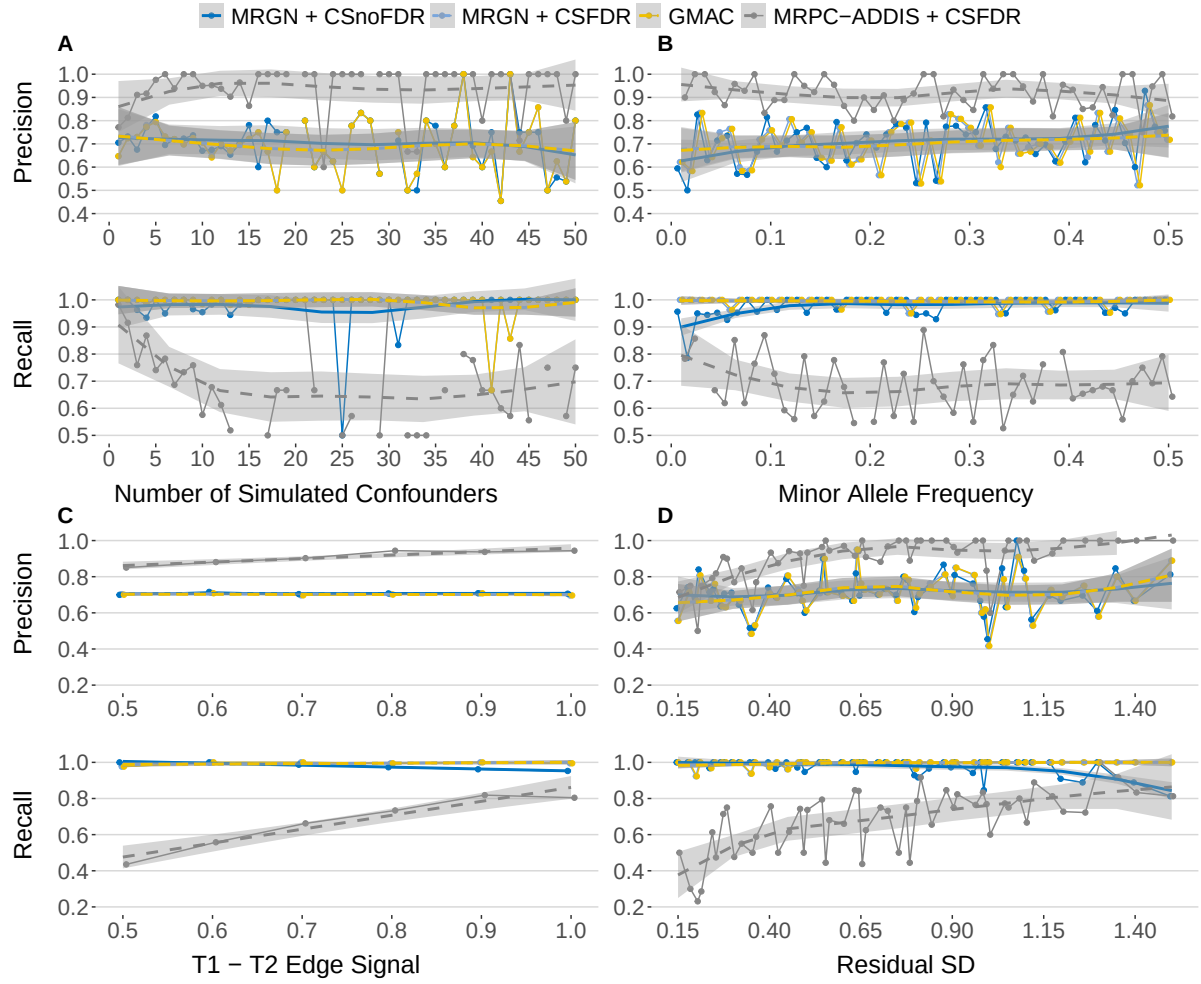
Test Type	STMS	LTMS	CMS	SIMS
Permutation test (Type II error)	0.1302	0.1302	0.1250	0.3125
Parametric test (Type II error)	0.0729	0.0729	0.0729	0.2969

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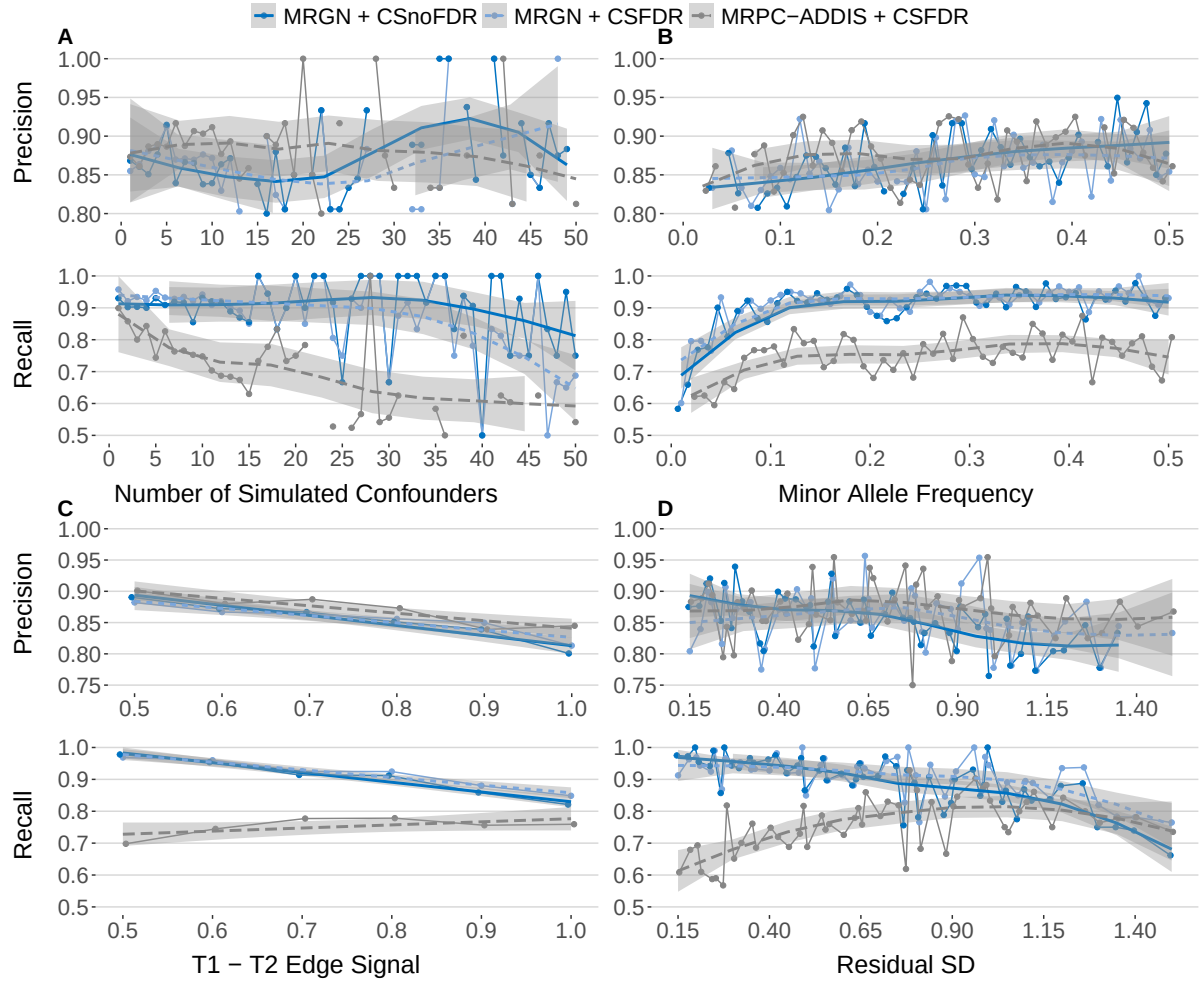
Supplementary Material



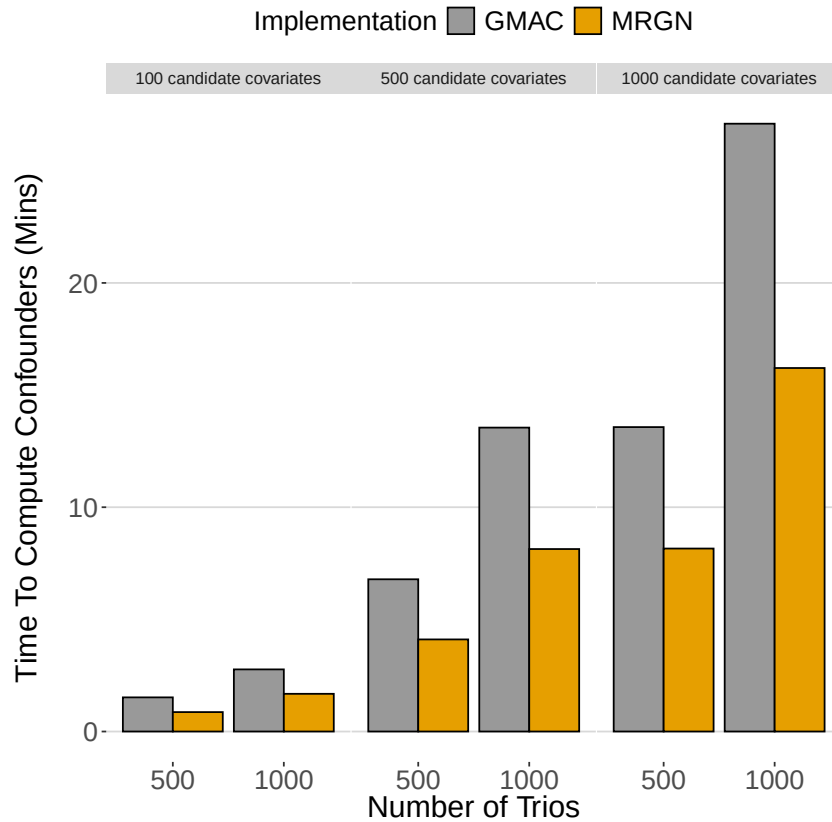
Supplementary Figure 1: The three types of confounding variables to consider in causal inference of genetic trios. A confounder (left) is a gene which regulates both molecular phenotypes, an intermediate variable (center) is a gene which mediates part of the effect of the genetic variant and its target, and a common child (right) is a shared target of both molecular phenotypes.



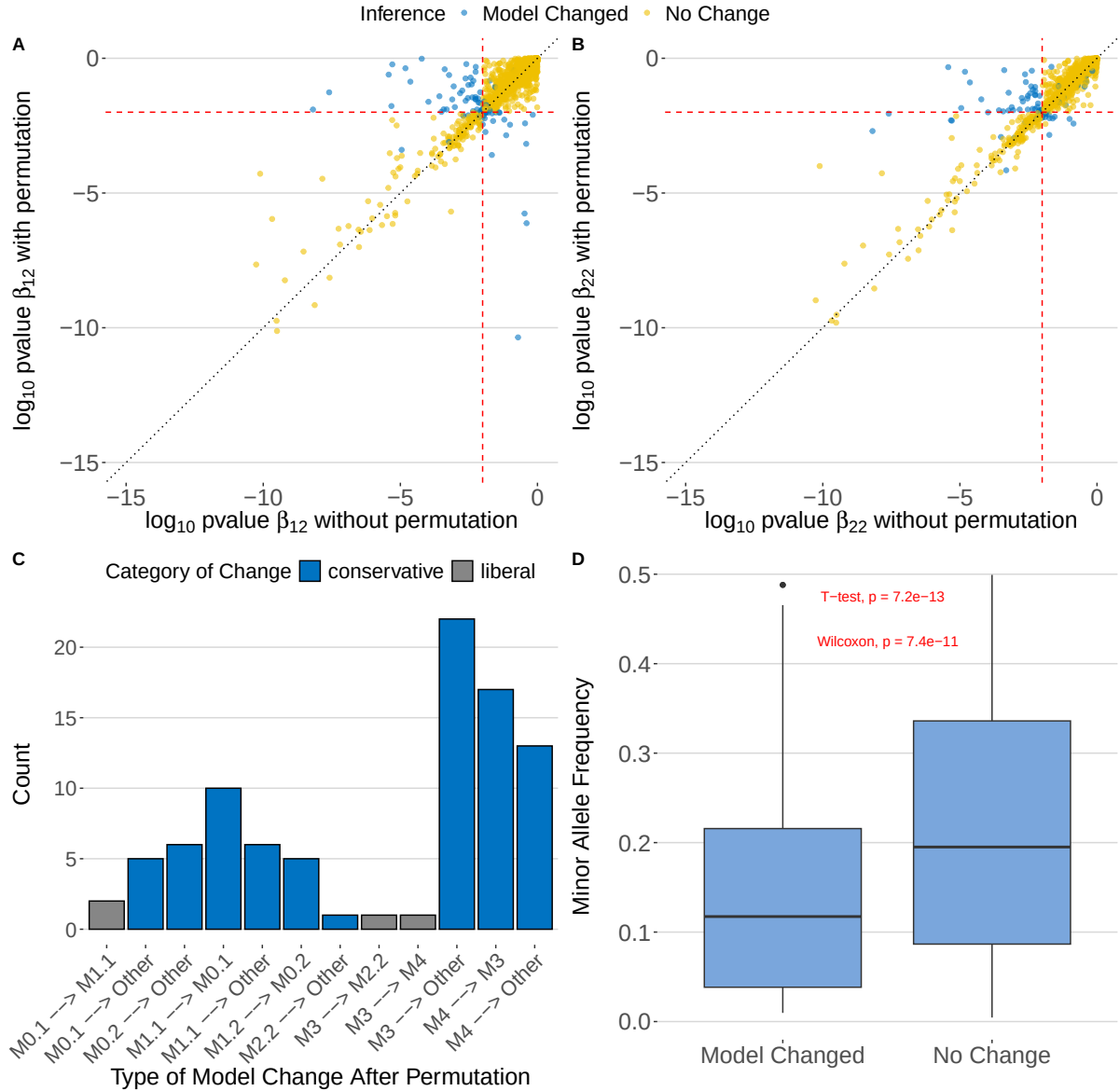
Supplementary Figure 2: Precision-recall curves for predicting the $T_1 - T_2$ edge across multiple simulation parameters. (A) the number of simulated hidden confounders (B) Minor allele frequency of the genetic variant, (C) The signal strength of the $T_1 - T_2$ edge, and (D) The standard deviation of the regression residuals.



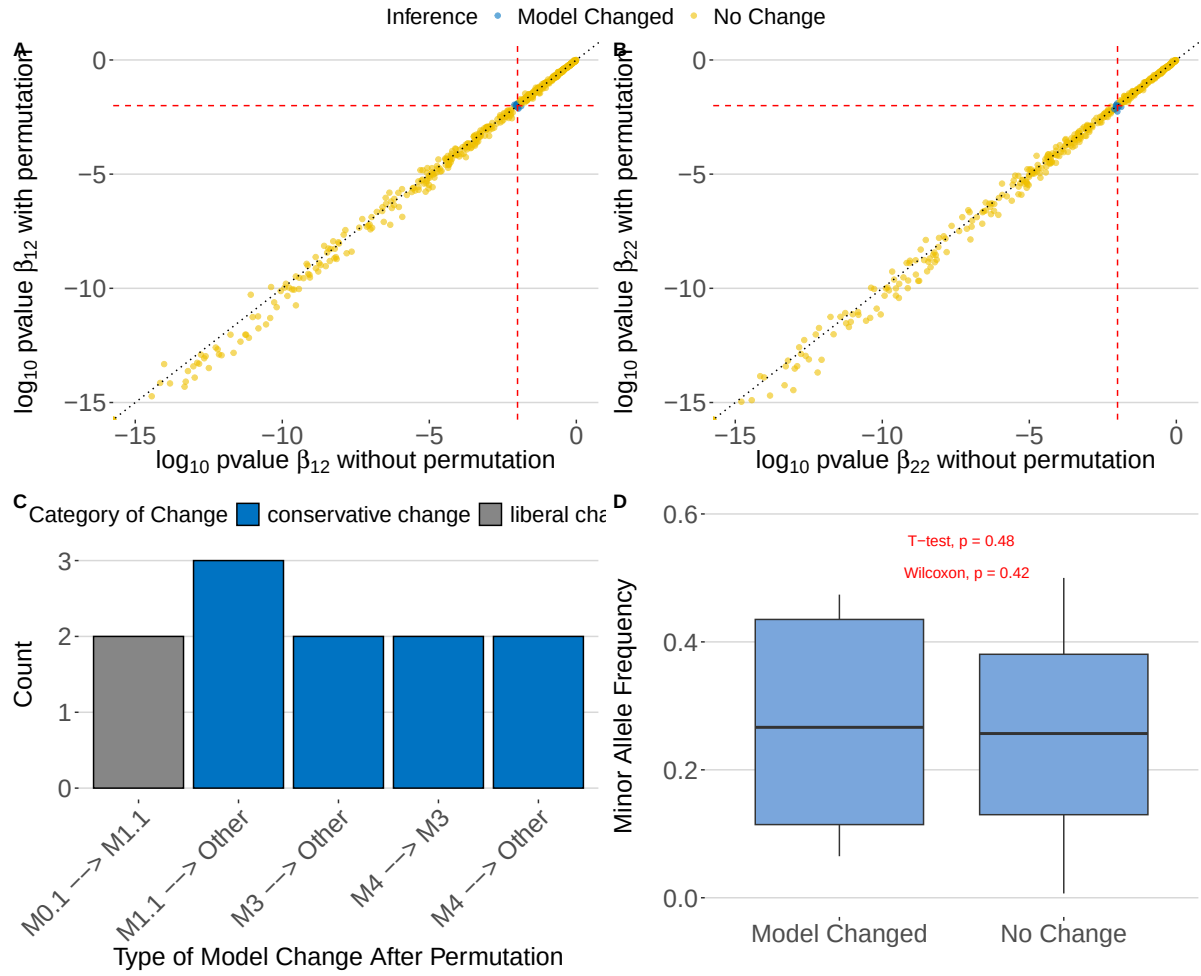
Supplementary Figure 3: Precision and recall curves of edge-based prediction performance for MRGNtrio, and MRPC with confounder selection. (A) the number of simulated hidden confounders (B) Minor allele frequency of the genetic variant, (C) The signal strength of the $T_1 - T_2$ edge, and (D) The standard deviation of the regression residuals.



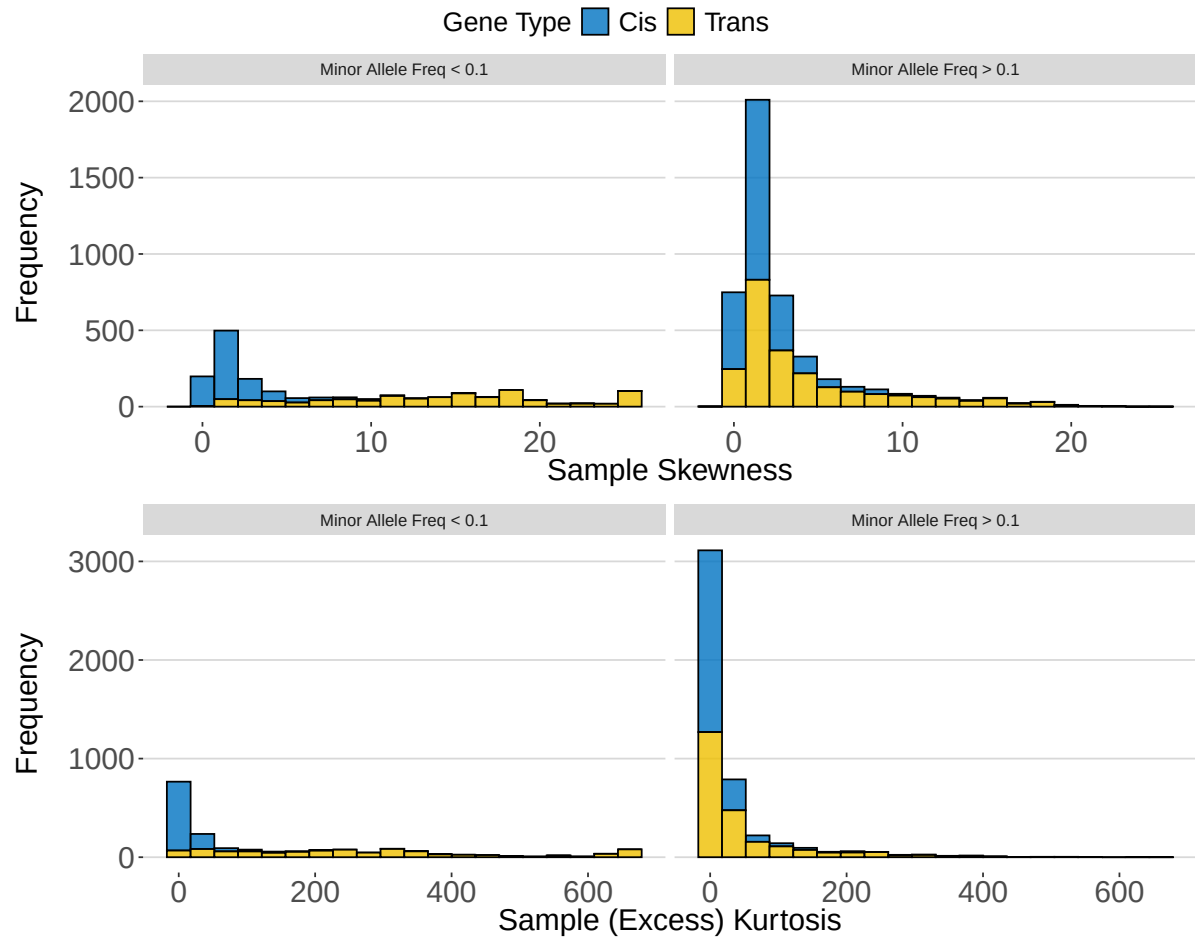
Supplementary Figure 4: Comparison of computation times between our implementation of the CSFDR method for confounder selection and that of the GMAC method [15]. Computation times are shown across different numbers of input trios and sizes of the candidate confounder pool.



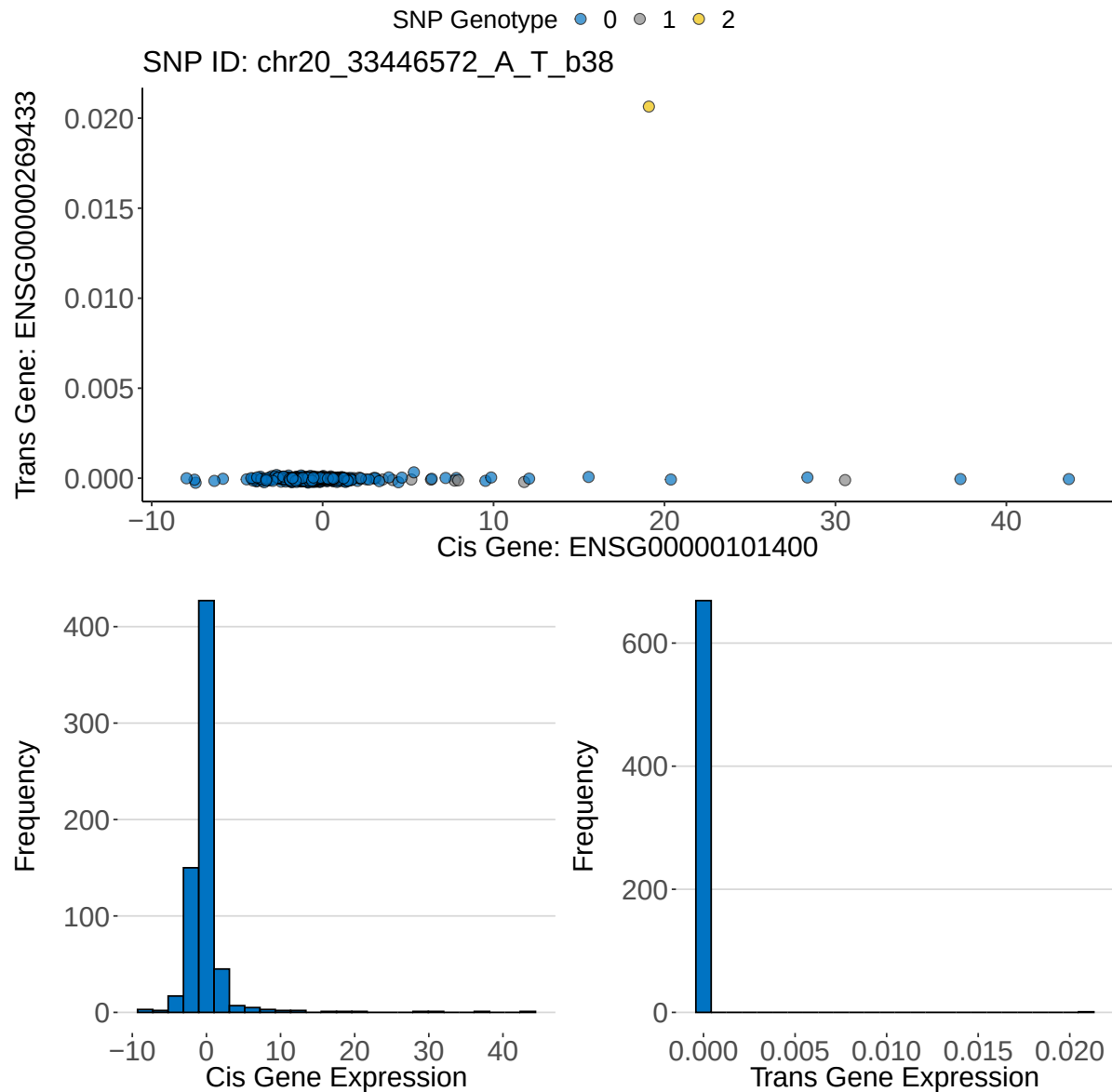
Supplementary Figure 5: Impact of the causal permutation test on inference and categorization of model changes for GTEx whole blood trios. (A) A p -value plot ($\log_{10}$ scale) comparing the observed p -value from the Wald test for $\beta_{12} : T_1 \perp\!\!\!\perp T_2 \mid (V, U)$ with the p -value from the permutation test. (B) A p -value plot for the conditional test $\beta_{22} : T_2 \perp\!\!\!\perp T_1 \mid (V, U)$ showing the permutation-based p -values (y-axis) versus the standard Wald test. Red dotted lines indicate the significance threshold on the log scale. (C) Categorization and counts of inferred model changes resulting from using the permutation test instead of the standard Wald test. (D) Minor allele frequencies of trios whose inferred models changed compared with those that did not change after applying the permutation test for β_{12} and β_{22} .



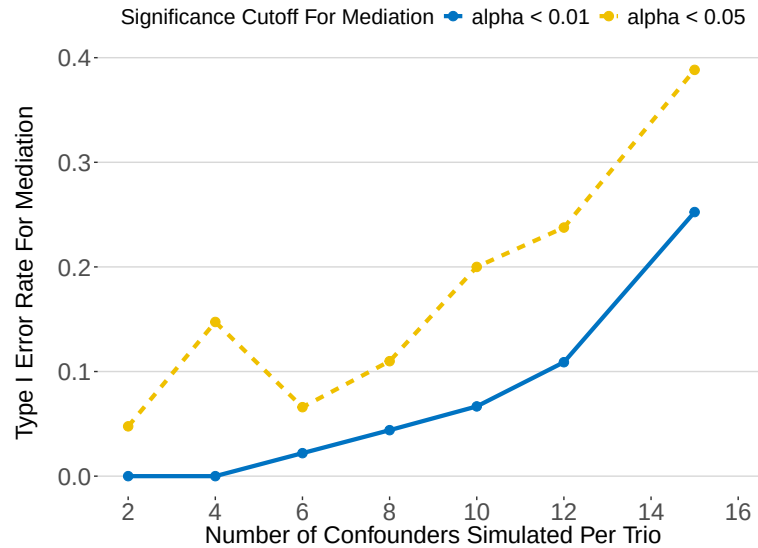
Supplementary Figure 6: Impact of the permutation test on inference and categorization of inferred model changes for simulated trios. (A) p -value plot ($\log_{10}$ scale) comparing the observed p -value from the Wald test for $\beta_{12} : T_1 \perp\!\!\!\perp T_2 \mid (V, U)$ with the p -value from the permutation test. (B) Same as (A) but for the conditional test $\beta_{22} : T_2 \perp\!\!\!\perp T_1 \mid (V, U)$. (C) Categorization and counts of inferred model changes resulting from using the permutation test instead of the standard Wald test. (D) Distribution of allele frequencies for trios whose inferred models changed using the permutation test compared with trios with no model change.



Supplementary Figure 7: Skewness and kurtosis for cis and trans genes by allele frequency. Stacked histograms showing the sampling distributions of sample skewness and excess kurtosis for cis and trans genes in GTEx trios. Results are segregated by the minor allele frequency of the genetic variant, highlighting differences between common and rare alleles.



Supplementary Figure 8: Example of a trio with a rare variant acting as an outlier. In this trio, a single individual homozygous for the minor allele drives non-normality in trans-gene expression relative to the population. This individual (yellow point) also shows elevated expression of the cis gene. Initially, MRGNTrio inferred model M4 for this trio, while GMAC inferred no mediation. Applying the permutation test caused MRGNTrio to recover the same inference (the null model) as GMAC.



Supplementary Figure 9: Results of simulations investigating type I error rate (false mediation) under the null model. Trios were simulated under the null model for each of seven scenarios: 2, 4, 6, 8, 10, 12 or 15 confounders, excluding common child and intermediate variables. The GMAC method was applied to trios in each scenario using CSFDR confounder selection.

650 Supplementary tables

Supplementary Table 1: A complete list of simulated trios, their simulation parameters, and inference results under MRGNtrio, MRPC-ADDIS, and GMAC.

Supplementary Table 2: All analyzed GTEx whole-blood trios, including genetic annotations (locus, variant ID, gene name, etc.) and inference results from MRGNtrio, MRPC-ADDIS, and GMAC.

Supplementary Table 3: This table compares MRGNtrio, MRPC, and GMAC on results from application to GTEx whole blood tissue. Results compared between MRGNtrio with CSnoFDR confounder selection to MRPC and GMAC.

Supplementary Text 1: Detailed supplementary text to explain the four simulation scenarios presented under the section “MRGNtrio is robust to model misspecification in causal mediation”. This text also includes supplementary examples with output illustrating differences in causal inference between MRGNtrio, GMAC, and MRPC on selected GTEx and simulated trios.